

RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.000419666$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, $p=0.000419666$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Tables

domain_retention tables

frame_domain_contingency_table

141	38
5	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	194	179	1	14	0.04988179669030733

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16

Partner_gene	Event_count
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2
RUFY2	1
RUFY3	2
SHROOM3	1
SLC4A4	1
SPECC1L	4
TFG	1
TIMM23B	3
TRIM24	1
UXS1	1

Per-event results (results.tsv)

event_id	sampl_e_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051781-T03-IM6-3	P-0051781-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0059857-T02-IM7-4	P-0059857-T02-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611615	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0065992-T01-IM7-5	P-0065992-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611698	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0049461-T02-IM6-6	P-0049461-T02-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611705	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 6379-T01-I M6-7	P-0036 379-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436114 18	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 7412-T01-I M6-8	P-0037 412-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436108 39	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 8611-T01-I M6-9	P-0048 611-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436112 37	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-006 4853-T01-I M7-10	P-0064 853-T0 1-IM7		CCDC6-R ET	CCDC6	out-of-frame	threesome	436095 11	10	627	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA
EVT-P-004 7144-T01-I M6-11	P-0047 144-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436118 88	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 8257-T01-I M6-12	P-0038 257-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436105 94	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-001 7365-T04-I M6-13	P-0017 365-T0 4-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436114 64	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 9548-T01-I M6-14	P-0039 548-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	436108 43	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 0111-T01-I M6-15	P-0040 111-T0 1-IM6		CCDC6-R ET	CCDC6	unknown	threesome	436099 67	11	640	False	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 1192-T01-I M6-16	P-0041 192-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610571	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T04-I M6-17	P-0022 642-T0 4-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T03-I M6-18	P-0022 642-T0 3-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-005 1781-T01-I M6-19	P-0051 781-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 2417-T01-I M6-20	P-0042 417-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610397	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 4901-T01-I M5-21	P-0004 901-T0 1-IM5		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611729	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RET inv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-002 2380-T03-I M6-22	P-0022 380-T0 3-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-006 8500-T01-I M7-23	P-0068 500-T0 1-IM7		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610322	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RET inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-005 7735-T01-I M6-24	P-0057 735-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611887	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RET inv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 8053-T01-I M6-26	P-0038 053-T0 1-IM6		CLIP1-RET	CLIP1	unknown	threonine_prime	43611121	12	713	True	retained	1.0	False	Protein kinase domain			CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512) Protein Fusion: mid-exon {CLIP1:RET}	NA
EVT-P-000 0252-T01-I M3-27	P-0000 252-T0 1-IM3		CUBN-RET	CUBN	unknown	threonine_prime	43610494	11	712	True	retained	1.0	False	Protein kinase domain			NA Antisense fusion	NA
EVT-P-003 8256-T01-I M6-28	P-0038 256-T0 1-IM6		ERC1-RET	ERC1	unknown	threonine_prime	43609964	11	639	False	retained	1.0	False	Protein kinase domain			ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1;p13.32)(chr10:g.43609964::chr12:g.1264106) Protein Fusion: mid-exon {ERC1:RET}	NA
EVT-P-003 4781-T01-I M6-29	P-0034 781-T0 1-IM6		GRIPAP1-RET	GRIPAP1	in-frame	threonine_prime	43611853	12	713	True	retained	1.0	False	Protein kinase domain			GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853) Protein Fusion: in frame {GRIPAP1:RET}	NA
EVT-P-002 6245-T01-I M6-30	P-0026 245-T0 1-IM6		KIAA1217-RET	KIAA1217	unknown	threonine_prime	43606666	7	425	False	retained	1.0	False	Protein kinase domain			KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel Protein Fusion: mid-exon {KIAA1217:RET}	NA
EVT-P-004 2153-T01-I M6-31	P-0042 153-T0 1-IM6		KIF13A-RET	KIF13A	in-frame	threonine_prime	43610510	11	712	True	retained	1.0	False	Protein kinase domain			KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510) Protein Fusion: in frame {KIF13A:RET}	NA
EVT-P-001 9535-T02-I M6-32	P-0019 535-T0 2-IM6		KIF5B-RET	KIF5B	in-frame	threonine_prime	43611913	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9257-T03-I M7-33	P-0039 257-T0 3-IM7		KIF5B-RET	KIF5B	unknown	threonine_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 9514-T01-I M5-34	P-0009 514-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threep _prime	436118 11	12	713	True	retained	1.0	False	Protein kinase domain			Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-005 5183-T01-I M6-35	P-0055 183-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436103 08	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 1035-T01-I M5-36	P-0011 035-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threep _prime	436113 22	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-001 2648-T01-I M5-37	P-0012 648-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threep _prime	436116 97	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-005 3175-T01-I M6-38	P-0053 175-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436111 62	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 2861-T02-I M6-39	P-0052 861-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436102 60	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 3825-T01-I M5-40	P-0013 825-T0 1-IM5		KIF5 B-RET	KIF5B	unknown	threep _prime	436115 20	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-005 2826-T01-I M6-41	P-0052 826-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436111 82	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 4808-T01-I M6-42	P-0014 808-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611055	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-003 4357-T01-I M6-43	P-0034 357-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611187	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 4808-T02-I M6-44	P-0014 808-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611056	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 3700-T02-I M6-45	P-0033 700-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611670	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 4808-T03-I M6-46	P-0014 808-T0 3-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611056	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 8289-T01-I M5-47	P-0008 289-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threesome	43611748	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 0252-T01-I M3-48	P-0000 252-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	threesome	43610597	11	712	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 8101-T01-I M5-49	P-0008 101-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threesome	43610455	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 8968-T01-I M6-50	P-0048 968-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43611809	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 8829-T01-I M6-51	P-0048 829-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43610616	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 5876-T01-I M7-52	P-0065 876-T0 1-IM7		KIF5 B-RET	KIF 5B	in-frame	threesome	43611299	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 6221-T01-I M6-53	P-0016 221-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43611268	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 7367-T01-I M6-54	P-0037 367-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43611874	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 7072-T02-I M6-55	P-0057 072-T0 2-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43611106	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 7214-T01-I M6-56	P-0017 214-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43610647	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 7214-T02-I M6-57	P-0017 214-T0 2-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43610647	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2753-T01-I M6-58	P-0032 753-T0 1-IM6		KIF5 B-RET	KIF 5B	in-frame	threesome	43610925	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 7072-T03-I M7-59	P-0057 072-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	threesome	43611106	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2686-T01-I M6-60	P-0032 686-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611982	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 7523-T02-I M6-61	P-0017 523-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611192	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 6827-T02-I M6-62	P-0046 827-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43610889	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2407-T01-I M6-63	P-0032 407-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611730	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 8069-T01-I M6-64	P-0018 069-T0 1-IM6		KIF5 B-RET	KIF5B	out-of-frame	five_prime	43607832	8	550	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv Protein Fusion: out of frame {RET:KIF5B}	NA
EVT-P-000 5573-T01-I M5-65	P-0005 573-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threesome	43610418	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 4995-T01-I M5-66	P-0004 995-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threesome	43610660	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B(NM_004521) -RET (NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-003 6092-T01-I M6-67	P-0036 092-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43610413	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 6726-T01-I M6-68	P-0046 726-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43610987	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2227-T01-I M6-69	P-0032 227-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43611125	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 0273-T02-I M6-70	P-0020 273-T0 2-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43611819	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 3007-T02-I M6-71	P-0043 007-T0 2-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43610813	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1150-T01-I M6-72	P-0021 150-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43611591	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1578-T01-I M6-73	P-0021 578-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43610809	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1879-T01-I M6-74	P-0021 879-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43610699	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1937-T01-I M6-75	P-0021 937-T0 1-IM6		KIF5B-RET	KIF5B	in-frame	threesome_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 1937-T03-I M6-76	P-0021 937-T0 3-IM6		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436115 55	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 3007-T01-I M6-77	P-0043 007-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436108 13	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 2608-T01-I M6-78	P-0042 608-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436118 02	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 2580-T01-I M6-79	P-0042 580-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436109 44	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 9913-T01-I M7-80	P-0059 913-T0 1-IM7		KIF5 B-RE T	KIF 5B	unknown	thre e_ pr ime	436101 08	11	687	False	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-000 1010-T02-I M5-81	P-0001 010-T0 2-IM5		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436108 63	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-006 0748-T02-I M7-82	P-0060 748-T0 2-IM7		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436115 27	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 2104-T01-I M6-83	P-0042 104-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436105 05	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 3125-T01-I M5-84	P-0003 125-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-frame	thre e_ pr ime	436115 84	12	713	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 3125-T01-I M5-85	P-0003 125-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-frame	five_ pr ime	436115 81	12	713	True	lost	0.0	False		Protein kinase domain		NA Protein fusion: in frame (RET-KIF5B)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 2873-T01-I M3-86	P-0002 873-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-002 2866-T01-I M6-87	P-0022 866-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 4225-T01-I M6-88	P-0024 225-T0 1-IM6		KIF5 B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-000 2694-T02-I M5-89	P-0002 694-T0 2-IM5		KIF5 B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-002 4843-T01-I M6-90	P-0024 843-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5100-T01-I M6-91	P-0025 100-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5100-T02-I M6-92	P-0025 100-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9941-T01-I M6-93	P-0039 941-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611339	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 0748-T03-I M7-94	P-0060 748-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9474-T02-I M6-95	P-0039 474-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9474-T01-I M6-96	P-0039 474-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 5183-T02-I M6-97	P-0055 183-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5838-T01-I M6-98	P-0025 838-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610972	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5928-T01-I M6-99	P-0025 928-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5928-T02-I M6-100	P-0025 928-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 6637-T01-I M6-101	P-0036 637-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9257-T01-I M6-102	P-0039 257-T0 1-IM6		KIF5 B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 6693-T01-I M6-103	P-0026 693-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610511	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RETinv. Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 6777-T01-I M6-104	P-0026 777-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 6777-T02-I M6-105	P-0026 777-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 8961-T01-I M6-106	P-0038 961-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610775	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 7723-T01-I M6-107	P-0027 723-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-003 6737-T02-I M6-108	P-0036 737-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 1331-T01-I M7-109	P-0061 331-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610884	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 7991-T01-I M6-110	P-0027 991-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 2948-T03-I M7-111	P-0062 948-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	threesome	43610211	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 4951-T01-I M7-112	P-0064 951-T0 1-IM7		KIF5 B-RET	KIF5B	out-of-frame	threesome	43611130	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv Protein Fusion: out of frame {KIF5B:RET}	NA
EVT-P-003 0453-T01-I M6-113	P-0030 453-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43610911	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 7494-T01-I M6-114	P-0037 494-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611755	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 0208-T02-I M5-115	P-0000 208-T0 2-IM5		KIF5 B-RET	KIF5B	in-frame	threesome	43611647	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 0208-T01-I M3-116	P-0000 208-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	threesome	43611647	12	713	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-001 6221-T02-I M6-117	P-0016 221-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome	43611268	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 9663-T01-I M6-118	P-0019 663-T0 1-IM6		LIN5 2-RET	LIN52	out-of-frame	fiveprime	43608613	9	587	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - LIN52 (NM_001024674) Rearrangement : t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750) Protein Fusion: out of frame {RET:LIN52}	NA
EVT-P-006 5826-T01-I M7-120	P-0065 826-T0 1-IM7		NCOA4-RET	NCOA4	out-of-frame	threesome	43610857	11	712	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 7961-T01-I M6-121	P-0037 961-T0 1-IM6		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43611448	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-003 8315-T01-I M6-122	P-0038 315-T0 1-IM6		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43611981	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-006 0765-T02-I M7-123	P-0060 765-T0 2-IM7		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43611844	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448:NCOA4dup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-004 3077-T01-I M6-124	P-0043 077-T0 1-IM6		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43611580	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-000 5748-T01-I M5-125	P-0005 748-T0 1-IM5		NCOA4-R ET	NC OA4	in-frame	threonine_prime	43611325	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-004 7991-T02-I M6-126	P-0047 991-T0 2-IM6		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43611538	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-004 8805-T01-I M6-127	P-0048 805-T0 1-IM6		NCOA4-R ET	NC OA4	unknown	threonine_prime	43611225	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-004 9600-T01-I M6-128	P-0049 600-T0 1-IM6		NCOA4-R ET	NC OA4	unknown	threonine_prime	43611613	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-005 6870-T01-I M6-129	P-0056 870-T0 1-IM6		NCOA4-R ET	NC OA4	out-of-frame	threonine_prime	43610337	11	712	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_n_cated	retain_ed_do_mains	lost_do_mai_ns	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 2409-T01-I M6-130	P-0052 409-T0 1-IM6		NCO A4-R ET	NC OA4	out- of-fr ame	thr ee _pr im e	436116 91	12	713	True	retain ed	1.0	False	Protei n kinase domai n			NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NC OA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-000 4094-T02-I M6-131	P-0004 094-T0 2-IM6		REL CH-R ET	REL CH	in-fr ame	thr ee _pr im e	436102 79	11	712	True	retain ed	1.0	False	Protei n kinase domai n			KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;1 8)(q11.21;q21.33)(chr10:g.436 10279::chr18:g.59909399) Protein Fusion: in frame {KIAA1468:RET}	NA
EVT-P-001 5588-T01-I M6-132	P-0015 588-T0 1-IM6		RET- CCD C186	CC DC1 86	in-fr ame	thr ee _pr im e	436103 08	11	712	True	retain ed	1.0	False	Protei n kinase domai n			C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142: C10orf118_c.2136+124:RETinv v Protein fusion: in frame (C10orf118-RET)	NA
EVT-P-002 0974-T01-I M6-133	P-0020 974-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr ee _pr im e	436111 84	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-8 48:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 1745-T02-I M5-134	P-0001 745-T0 2-IM5		RET- CCD C6	CC DC6	in-fr ame	thr ee _pr im e	436112 45	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-7 87:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-001 7365-T02-I M6-135	P-0017 365-T0 2-IM6		RET- CCD C6	CC DC6	in-fr ame	thr ee _pr im e	436114 64	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_ c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 1898-T01-I M3-136	P-0001 898-T0 1-IM3		RET- CCD C6	CC DC6	in-fr ame	thr ee _pr im e	436116 99	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-002 9602-T01-I M6-137	P-0029 602-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr ee _pr im e	436111 60	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-57 68:CCDC6_c.2137-872:RETinv v Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004916-T01-I M5-138	P-0004916-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0000425-T01-I M3-139	P-0000425-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	lost	0.0	False		Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0031330-T01-I M6-140	P-0031330-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20) : c.303+3703:CCDC6_c.2136+169:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0030990-T03-I M6-141	P-0030990-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611881	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0005168-T01-I M5-142	P-0005168-T01-IM5		RET-CCDC6	CCDC6	out-of-frame	three_prime	43608797	10	587	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv Protein fusion: out of frame (CCDC6-RET)	NA
EVT-P-0022380-T01-I M6-143	P-0022380-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20) : c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022380-T02-I M6-144	P-0022380-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 9315-T02-I M6-145	P-0029 315-T0 2-IM6		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436106 31	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 9315-T01-I M6-146	P-0029 315-T0 1-IM6		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436106 31	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 0889-T01-I M3-147	P-0000 889-T0 1-IM3		RET- CCD C6	CC DC6	in-frame	five _pr _im _e	436119 94	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-000 7322-T02-I M6-148	P-0007 322-T0 2-IM6		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436102 68	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 7953-T01-I M6-149	P-0027 953-T0 1-IM6		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436112 78	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 8504-T03-I M6-150	P-0008 504-T0 3-IM6		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436105 83	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 8955-T03-I M5-151	P-0008 955-T0 3-IM5		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436110 19	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-001 3023-T01-I M5-152	P-0013 023-T0 1-IM5		RET- CCD C6	CC DC6	in-frame	thre e_ _pr _im _e	436118 10	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET)	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_p_o_int	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_n_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 3394-T01-I M5-153	P-0013 394-T0 1-IM5		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436102 32	11	712	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RET inv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-002 2607-T01-I M6-154	P-0022 607-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436112 12	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T01-I M6-155	P-0022 642-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436117 68	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T02-I M6-156	P-0022 642-T0 2-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436117 68	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-001 7365-T03-I M6-157	P-0017 365-T0 3-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436114 64	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 5349-T03-I M6-158	P-0025 349-T0 3-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436119 86	12	713	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 9588-T01-I M6-159	P-0049 588-T0 1-IM6		RET- CCD C6	CC DC6	out-of-fr ame	five _pr im_e	436094 37	10	627	True	lost	0.0	False		Prot ein kina se d oma in		RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv Protein Fusion: out of frame {RET:CCDC6}	NA
EVT-P-001 5435-T01-I M6-160	P-0015 435-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436109 69	11	712	True	retain ed	1.0	False	Protei n kinase domai n			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0005556-T01-I M5-161	P-0005556-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0063502-T01-I M7-162	P-0063502-T01-IM7		RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	retained	1.0	False	Protein kinase domain			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038) Protein Fusion: mid-exon {RET:CTNNA1}	NA
EVT-P-0063802-T01-I M7-163	P-0063802-T01-IM7		RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	disrupted	0.74822695035461	True			Protein kinase domain	RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv Protein Fusion: out of frame {RET:CTNNA3}	NA
EVT-P-0030396-T02-I M6-164	P-0030396-T02-IM6		RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	retained	1.0	False	Protein kinase domain			DOCK1 (NM_001380) - RET (NM_020975)Rearrangement: c.2515+295:RET_c.1668:RETdup Protein Fusion: mid-exon {DOCK1:RET}	NA
EVT-P-0013886-T01-I M5-165	P-0013886-T01-IM5		RET-EML4	EML4	unknown	three_prime	43610067	11	673	False	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) Protein fusion: mid-exon (EML4-RET)	NA
EVT-P-0014808-T03-I M6-166	P-0014808-T03-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T02-I M6-167	P-0014808-T02-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T01-I M6-168	P-0014808-T01-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein fusion: out of frame (RET-ERCC6)	NA
EVT-P-0045809-T02-I M6-169	P-0045809-T02-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622269	19	1063	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269) Protein Fusion: in frame {RET:FBXL7}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 5809-T06-I M6-170	P-0045 809-T0 6-IM6		RET- FBXL 7	FBX L7	in-frame	five_prime	436223 23	19	1063	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323) Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-002 5215-T01-I M6-171	P-0025 215-T0 1-IM6		RET- GEMIN 5	GEMIN 5	in-frame	three_prime	436110 67	11	712	True	retained	1.0	False	Protein kinase domain			GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067) Protein Fusion: in frame {GEMIN5:RET}	NA
EVT-P-002 7417-T01-I M6-173	P-0027 417-T0 1-IM6		RET- KCNMA 1	KCNMA 1	out-of-frame	three_prime	436081 58	9	550	True	retained	1.0	False	Protein kinase domain			KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET Protein Fusion: out of frame {KCNMA1:RET}	NA
EVT-P-004 3704-T01-I M6-174	P-0043 704-T0 1-IM6		RET- KIF5B	KIF5B	in-frame	five_prime	436103 31	11	712	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv Protein Fusion: in frame {RET:KIF5B}	NA
EVT-P-000 6226-T02-I M5-178	P-0006 226-T0 2-IM5		RET- NCOA A4	NCOA A4	in-frame	three_prime	436118 93	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame {NCOA4-RET}	NA
EVT-P-000 1151-T01-I M3-179	P-0001 151-T0 1-IM3		RET- NCOA A4	NCOA A4	in-frame	three_prime	436116 45	12	713	True	retained	1.0	False	Protein kinase domain			RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup Protein fusion: in frame {NCOA4-RET}	NA
EVT-P-000 9592-T01-I M5-180	P-0009 592-T0 1-IM5		RET- NCOA A4	NCOA A4	unknown	three_prime	436111 43	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon {NCOA4-RET}	NA
EVT-P-000 1155-T01-I M3-181	P-0001 155-T0 1-IM3		RET- NCOA A4	NCOA A4	in-frame	three_prime	436109 41	11	712	True	retained	1.0	False	Protein kinase domain			RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup Protein fusion: in frame {NCOA4-RET}	NA
EVT-P-002 2818-T01-I M6-182	P-0022 818-T0 1-IM6		RET- NCOA A4	NCOA A4	unknown	three_prime	436117 46	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 2145-T01-I M3-183	P-0002 145-T0 1-IM3		RET- PDC D10	PDC D10	out- of-fr ame	five _pr ime	436093 94	10	627	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10;3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184) Protein fusion: out of frame (RET-PDCD10)	NA
EVT-P-000 1898-T01-I M3-184	P-0001 898-T0 1-IM3		RET- RAS SF4	RAS SF4	in-fr ame	five _pr ime	436117 49	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del Protein fusion: in frame (RET-RASSF4)	NA
EVT-P-003 0396-T02-I M6-185	P-0030 396-T0 2-IM6		RET- RBP MS	RBP MS	unk now n	three _pr ime	436100 18	11	657	False	retained	1.0	False	Protein kinase domain			RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8;10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018) Protein Fusion: mid-exon {RBPMS:RET}	NA
EVT-P-002 9222-T02-I M6-212	P-0029 222-T0 2-IM6		RET- RUF Y1	RUF Y1	unk now n	three _pr ime	436100 37	11	663	False	retained	1.0	False	Protein kinase domain			RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5;10)(q35.3,q11.21)(chr5:g.179025890::chr10:g.43610037) Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-001 5310-T02-I M6-213	P-0015 310-T0 2-IM6		RET- RUF Y3	RUF Y3	out- of-fr ame	three _pr ime	436104 27	11	712	True	retained	1.0	False	Protein kinase domain			RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427) Protein fusion: out of frame (RUFY3-RET)	NA
EVT-P-001 5310-T04-I M6-214	P-0015 310-T0 4-IM6		RET- RUF Y3	RUF Y3	unk now n	three _pr ime	436104 27	11	712	True	retained	1.0	False	Protein kinase domain			RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655251::chr10:g.43610427) Protein Fusion: mid-exon {RUFY3:RET}	NA
EVT-P-002 5609-T01-I M6-215	P-0025 609-T0 1-IM6		RET- SHR OOM 3	SHR OOM 3	in-fr ame	three _pr ime	436115 45	12	713	True	retained	1.0	False	Protein kinase domain			SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1;q11.21)(chr4:g.77664882::chr10:g.43611545) Protein Fusion: in frame {SHROOM3:RET}	NA
EVT-P-005 3530-T01-I M6-216	P-0053 530-T0 1-IM6		RET- SLC4 A4	SLC 4A4	out- of-fr ame	five _pr ime	436104 39	11	712	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.72412310::chr10:g.43610439) Protein Fusion: out of frame {RET:SLC4A4}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 2145-T01-I M3-217	P-0002 145-T0 1-IM3		RET-TFG	TFG	in-frame	threonine_prime	43609437	10	627	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - TFG (NM_001195478) translocation: t(10;3)(q11.2;q12.2)(chr10:g.43609437::chr3:g.100450751) Protein fusion: in frame (TFG-RET)	NA
EVT-P-002 1980-T01-I M6-218	P-0021 980-T0 1-IM6		RET-TIMM23B	TIMM23B	out-of-frame	threonine_prime	43610550	11	712	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12- 20): c.763-591:NCOA4_c.2136+366:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-001 7716-T01-I M6-219	P-0017 716-T0 1-IM6		RET-TIMM23B	TIMM23B	out-of-frame	threonine_prime	43611172	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-002 5531-T01-I M6-220	P-0025 531-T0 1-IM6		RET-TIMM23B	TIMM23B	out-of-frame	threonine_prime	43611375	12	713	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108NCOA4_c.2137-657RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-003 4704-T01-I M6-222	P-0034 704-T0 1-IM6		RET-TRIM24	TRIM24	in-frame	threonine_prime	43611827	12	713	True	retained	1.0	False	Protein kinase domain			TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43611827) Protein Fusion: in frame {TRIM24:RET}	NA
EVT-P-004 6790-T01-I M6-223	P-0046 790-T0 1-IM6		RET-UXS1	UXS1	unknown	five_prime	43600430	4	219	False	lost	0.0	False		Protein kinase domain		RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430) Protein Fusion: mid-exon {RET:UXS1}	NA
EVT-P-002 9222-T03-I M6-224	P-0029 222-T0 3-IM6		RUFY1-RET	RUFY1	unknown	threonine_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain			RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037) Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-003 1340-T02-I M6-225	P-0031 340-T0 2-IM6		RUFY2-RET	RUFY2	in-frame	threonine_prime	43611559	12	713	True	retained	1.0	False	Protein kinase domain			RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv Protein Fusion: in frame {RUFY2:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 6614-T01-I M6-227	P-0026 614-T0 1-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43611575	12	713	True	retained	1.0	False	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T06-I M6-228	P-0036 400-T0 6-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T04-I M6-229	P-0036 400-T0 4-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T03-I M6-230	P-0036 400-T0 3-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA

Figures

domain retention outliers

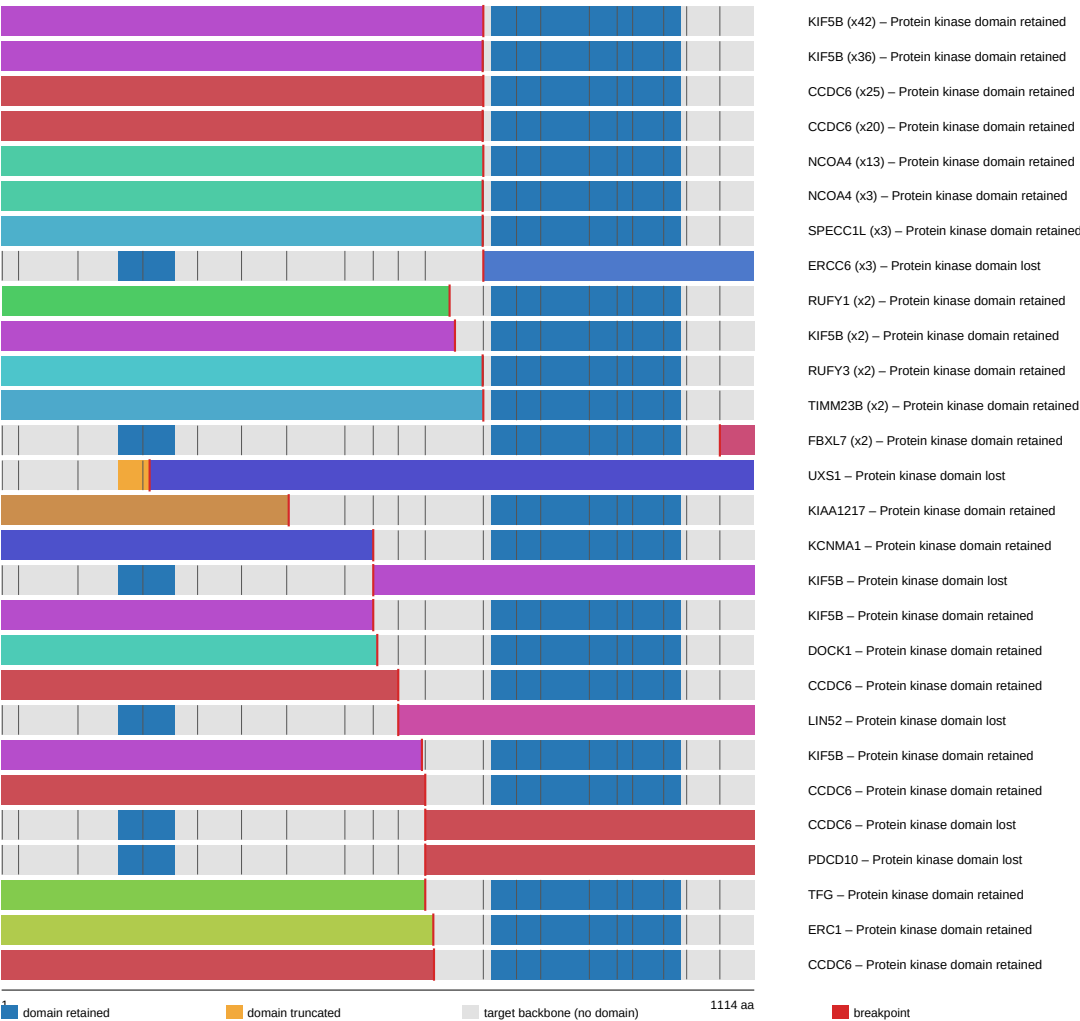
RET domain-retention track



fusion schematic

RET fusion-transcript schematic

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)



Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

